

JOB SHEET 13 – jQuery Project

Student Task Manager

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Student Name		Date	
Project	Student Task Manager (own project)	Marks	

1. Objective

Build an original web application using jQuery without a supplied question. The project applies selectors, events, effects, DOM manipulation, utility functions and browser storage in one working page.

2. Project Description

Student Task Manager lets a student add study tasks with a priority, tick them as done, delete them, search, filter by status and switch between light and dark mode. Tasks are saved in the browser so they remain after the page is reloaded.

3. Files

File	Purpose
index.html	Page structure: input area, search, filter buttons, task list, counter
style.css	Layout, priority colours, completed-task style and dark theme
script.js	All jQuery code for the application
jquery.js	jQuery library (3.6.1)

4. Features and jQuery Used

Feature	jQuery used
Run code when page is ready	<code>\$(document).ready()</code>
Add a task (button or Enter key)	<code>click()</code> , <code>keypress()</code> , <code>val()</code> , <code>\$.trim()</code> , <code>focus()</code>
Build each list item	<code>\$("")</code> , <code>attr()</code> , <code>addClass()</code> , <code>toggleClass()</code> , <code>text()</code> , <code>appendTo()</code> , <code>append()</code> , <code>empty()</code>
Mark task as done / delete	<code>on()</code> with event delegation, <code>closest()</code> , <code>data()</code> , <code>prop()</code>
Animations	<code>slideDown()</code> , <code>slideUp()</code> , <code>fadeIn()</code> , <code>fadeOut()</code> , <code>delay()</code> , <code>stop()</code>
Filter All / Active / Done	<code>removeClass()</code> , <code>addClass()</code> , <code>data("filter")</code> , <code>each()</code> , <code>toggle()</code>
Live search	<code>keyup()</code> , <code>find()</code> , <code>toLowerCase()</code>
Counters and data handling	<code>\$.each()</code> , <code>\$.grep()</code> , <code>text()</code>
Dark / light mode	<code>toggleClass()</code> , <code>hasClass()</code>
Save tasks in the browser	<code>localStorage</code> with <code>JSON.stringify()</code> / <code>JSON.parse()</code>

5. Steps

1. Create a project folder named **jobsheet13-task-manager** and copy `jquery.js` into it.
2. Create **index.html** and link `style.css`, `jquery.js` and `script.js` inside `<head>`.
3. Add the input box, priority list, Add Task button, search box, filter buttons, task list and counter.
4. Create **style.css** for layout, priority border colours, the `.done` style and the `.dark` theme.
5. Create **script.js**. Start with `$(document).ready()` and write the `add`, `complete`, `delete`, `filter`, `search` and `clear` functions.
6. Save tasks with `localStorage` and load them when the page opens.
7. Open `index.html` in the browser and test every feature using the checklist below.

6. How to Run

This project does not need a web server. Double-click **index.html** to open it in Chrome or Edge. Keep all four files in the same folder.

7. Source Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Task Manager - jQuery Project</title>
  <link rel="stylesheet" href="style.css">
  <script src="jquery.js"></script>
  <script src="script.js"></script>
</head>
<body>
  <div id="app">
    <header>
      <h1>Student Task Manager</h1>
      <button id="themeBtn" type="button">Dark Mode</button>
    </header>

    <section id="addBox">
      <input type="text" id="taskInput" placeholder="Enter a new task..." maxlength="80">
      <select id="priority">
        <option value="low">Low</option>
        <option value="medium" selected>Medium</option>
        <option value="high">High</option>
      </select>
      <button id="addBtn" type="button">Add Task</button>
      <p id="message"></p>
    </section>

    <section id="tools">
      <input type="text" id="searchInput" placeholder="Search tasks...">
      <div id="filters">
        <button type="button" class="filterBtn active" data-filter="all">All</button>
        <button type="button" class="filterBtn" data-filter="active">Active</button>
        <button type="button" class="filterBtn" data-filter="done">Done</button>
      </div>
    </section>

    <ul id="taskList"></ul>
    <p id="emptyMsg">No tasks to show.</p>

    <footer>
      <span id="counter"></span>
      <button id="clearDone" type="button">Clear Completed</button>
    </footer>
  </div>
</body>
</html>
```

style.css

```
* { box-sizing: border-box; }
body {
  margin: 0;
  padding: 30px 12px;
  font-family: Arial, Helvetica, sans-serif;
  background: #eef2f7;
  color: #222;
}
body.dark { background: #14181f; color: #e8ecf1; }

#app {
  max-width: 640px;
  margin: 0 auto;
  background: #fff;
  border-radius: 12px;
  padding: 22px;
  box-shadow: 0 6px 20px rgba(0, 0, 0, .12);
}
body.dark #app { background: #1f2530; }

header { display: flex; justify-content: space-between; align-items: center; }
h1 { margin: 0; font-size: 1.6em; color: #2d5bd1; }
body.dark h1 { color: #8fb0ff; }

button {
  cursor: pointer;
  border: 0;
  border-radius: 6px;
  padding: 9px 14px;
  background: #2d5bd1;
  color: #fff;
  font-size: 14px;
}
button:hover { opacity: .88; }

input[type="text"], select {
  padding: 9px;
  border: 1px solid #b9c2d0;
  border-radius: 6px;
  font-size: 14px;
}
body.dark input[type="text"], body.dark select {
  background: #2a3140;
  color: #e8ecf1;
  border-color: #424c60;
}

#addBox { margin-top: 18px; display: flex; gap: 8px; flex-wrap: wrap; }
#taskInput { flex: 1; min-width: 180px; }
#message { width: 100%; margin: 0; min-height: 18px; font-size: 13px; color: #c62828; }

#tools {
  margin-top: 12px;
  display: flex;
  gap: 10px;
  flex-wrap: wrap;
  align-items: center;
  justify-content: space-between;
}
#searchInput { flex: 1; min-width: 160px; }
.filterBtn { background: #dfe6f3; color: #333; }
.filterBtn.active { background: #2d5bd1; color: #fff; }

#taskList { list-style: none; padding: 0; margin: 16px 0 0; }
#taskList li {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 11px 12px;
}
```

```

    margin-bottom: 8px;
    border-radius: 8px;
    background: #f5f7fb;
    border-left: 6px solid #9aa5b8;
}
body.dark #taskList li { background: #2a3140; }
#taskList li.low { border-left-color: #43a047; }
#taskList li.medium { border-left-color: #fb8c00; }
#taskList li.high { border-left-color: #e53935; }
#taskList li .text { flex: 1; word-break: break-word; }
#taskList li.done .text { text-decoration: line-through; opacity: .55; }
#taskList li .tag { font-size: 11px; text-transform: uppercase; opacity: .7; }
#taskList li .del { background: #e53935; padding: 5px 10px; }
#taskList li.found { outline: 2px solid #fdd835; }

#emptyMsg { text-align: center; color: #8893a5; display: none; }

footer {
    margin-top: 14px;
    display: flex;
    justify-content: space-between;
    align-items: center;
    font-size: 14px;
}
#clearDone { background: #6b7385; }

```

script.js

```
// Student Task Manager - jQuery Project
$(document).ready(function () {

    var STORE_KEY = "jobsheet13Tasks";
    var tasks = []; // { id, text, priority, done }
    var currentFilter = "all";

    /* ----- Storage (safe: works even if storage is blocked) ----- */
    function save() {
        try { localStorage.setItem(STORE_KEY, JSON.stringify(tasks)); } catch (e) { }
    }
    function load() {
        try {
            var data = localStorage.getItem(STORE_KEY);
            if (data) { tasks = JSON.parse(data); return; }
        } catch (e) { }
        tasks = [
            { id: 1, text: "Finish jQuery Job Sheet 13", priority: "high", done: false },
            { id: 2, text: "Revise HTML and CSS notes", priority: "medium", done: false },
            { id: 3, text: "Submit assignment to lecturer", priority: "low", done: true }
        ];
    }

    /* ----- Rendering ----- */
    function render() {
        var $list = $("#taskList").empty();
        $.each(tasks, function (i, t) {
            var $li = $("- </li>")
                .attr("data-id", t.id)
                .addClass(t.priority)
                .toggleClass("done", t.done);
            $("<input type='checkbox' class='chk'>").prop("checked", t.done).appendTo($li);
            $("<span class='text'></span>").text(t.text).appendTo($li);
            $("<span class='tag'></span>").text(t.priority).appendTo($li);
            $("<button type='button' class='del'>X</button>").appendTo($li);
            $list.append($li);
        });
        applyView();
    }

    function applyView() {
        var keyword = $.trim($("#searchInput").val()).toLowerCase();
        var visible = 0;
        $("#taskList li").each(function () {
            var $li = $(this);
            var isDone = $li.hasClass("done");
            var matchFilter = currentFilter === "all" ||
                (currentFilter === "done" && isDone) ||
                (currentFilter === "active" && !isDone);
            var text = $li.find(".text").text().toLowerCase();
            var matchSearch = keyword === "" || text.indexOf(keyword) !== -1;
            $li.toggle(matchFilter && matchSearch);
            if (matchFilter && matchSearch) { visible++; }
        });
        $("#emptyMsg").toggle(visible === 0);
        updateCounter();
    }

    function updateCounter() {
        var total = tasks.length;
        var done = $.grep(tasks, function (t) { return t.done; }).length;
        $("#counter").text(
            "Total: " + total + " | Active: " + (total - done) + " | Done: " + done
        );
    }

    function showMessage(text) {
        $("#message").text(text).stop(true, true).fadeIn(150).delay(1800).fadeOut(400);
    }
}

```

```

/* ----- Add task ----- */
function addTask() {
    var text = $.trim($("#taskInput").val());
    if (text === "") {
        showMessage("Please enter a task first.");
        $("#taskInput").focus();
        return;
    }
    var id = Date.now();
    tasks.push({ id: id, text: text, priority: $("#priority").val(), done: false });
    save();
    render();
    $("#taskList li[data-id='" + id + "']").hide().slideDown(250);
    $("#taskInput").val("").focus();
}

$("#addBtn").click(addTask);
$("#taskInput").keypress(function (e) {
    if (e.which === 13) { addTask(); }
});

/* ----- Complete / delete (event delegation) ----- */
$("#taskList").on("change", ".chk", function () {
    var id = $(this).closest("li").data("id");
    $.each(tasks, function (i, t) { if (t.id === id) { t.done = !t.done; } });
    $(this).closest("li").toggleClass("done");
    save();
    applyView();
});

$("#taskList").on("click", ".del", function () {
    var $li = $(this).closest("li");
    var id = $li.data("id");
    $li.slideUp(250, function () {
        tasks = $.grep(tasks, function (t) { return t.id !== id; });
        save();
        render();
    });
});

/* ----- Filter and search ----- */
$("#filters").on("click", ".filterBtn", function () {
    $(".filterBtn").removeClass("active");
    $(this).addClass("active");
    currentFilter = $(this).data("filter");
    applyView();
});

$("#searchInput").keyup(applyView);

/* ----- Clear completed ----- */
$("#clearDone").click(function () {
    var doneCount = $.grep(tasks, function (t) { return t.done; }).length;
    if (doneCount === 0) {
        showMessage("No completed tasks to clear.");
        return;
    }
    $("#taskList li.done").fadeOut(250, function () {
        tasks = $.grep(tasks, function (t) { return !t.done; });
        save();
        render();
    });
});

/* ----- Dark mode ----- */
$("#themeBtn").click(function () {
    $("body").toggleClass("dark");
    $(this).text($("body").hasClass("dark") ? "Light Mode" : "Dark Mode");
});
/* ----- Start ----- */

```

```
load();  
render();  
});
```

8. Testing Checklist

- [] Page opens with 3 sample tasks and the correct counter
- [] A new task is added with the chosen priority colour
- [] Pressing Enter in the input box adds a task
- [] Empty input shows a warning message
- [] Ticking a task strikes it through and updates Done count
- [] Delete button removes the task with a slide animation
- [] All / Active / Done filters show the correct tasks
- [] Search box filters tasks while typing
- [] Clear Completed removes only done tasks
- [] Dark Mode / Light Mode button switches the theme
- [] Tasks are still there after reloading the page